[Unique decomposition Theorem.
For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t
$$MB \text{ is unitary.}$$
Proof. Since B is invertible, the row vectors of B, $\{\beta_1, \cdots, \beta_n\}$ is basis for V.
Apply Gram-Schmidt process, that is, for each $1 \le k \le n$, put
$$\alpha_k = \beta_k - \sum_{k>j} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \cdot \alpha_j$$
Then, span $\{\alpha_1, \cdots, \alpha_r\}$ = span $\{\beta_1, \cdots, \beta_r\}$ and $\{\alpha_1, \cdots, \alpha_r\}$ $(r \le n)$ is orthogonal.
Put $U = \left[\frac{\alpha_1}{\|\alpha_1\|} \cdots \frac{\alpha_n}{\|\alpha_n\|}\right]$, then clearly $U^* \cdot U = I$, because $\{\alpha_1, \cdots, \alpha_n\}$ is orthogonal.
And, for each $1 \le k \le n$, $\alpha_k = \beta_k - \sum_{k>j} C_{kj} \beta_j$ for some C_{kj}.
Put
$$M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \frac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$$
Then, each $1 \le k \le n$, $U_k =$
$$\begin{pmatrix} \ddots & 0 \\ M_{kj} & \ddots \end{pmatrix} \cdot \begin{pmatrix} \beta_1 \cdots \beta_n \end{pmatrix} = U$$

So $MB = U$. To show the uniqueness, denote
$T^+(n) = \{M \in M_{n\times n}(\mathbb{C}) \mid M \text{ is lower triangle, positive diagonals.}\}$.
Let $M_1, M_2 \in T^+(n)$ satisfies $M_1 B$, $M_2 B$ are unitary. Since unitary matrices forms a group,
$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$
And, M_2^{-1} is lower-triangular with positive diagonals, so is $M_1 M_2^{-1}$. Now,
$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$
So, $M_1 M_2^{-1} = I$, or $M_1 = M_2$. □

[Corollary. For each $B \in GL(n)$, there exists unique matrix K, A, N s.t $B = KAN$,
where K is unitary, A is diagonal with positive values, N is upper-triangle with diagonals 1

†Note. Let $T_n(F)$ be the set of $n\times n$ upper-triangle matrices with all diagonals are 1.
Then, $(T_n(F), \cdot)$ is group, because: for any $A \in T_n(F)$, $A = I + N$,
where N is strictly upper triangular with $N^n = 0$, nilpotent. So,
$$A^{-1} = I - N + N^2 - N^3 + \cdots + (-N)^{n-1} \left(= \sum_{k=0}^{n-1} (-N)^k\right)$$
and $A^{-1} \in T_n(F)$ is clear.

QR - Factorization.

Let $A = (w_1 \ \cdots \ w_n)$ be an invertible matrix.

And, for each $1 \le k \le n$, $v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} \cdot v_i$.

Then, $\{v_1, \cdots, v_n\}$ is an orthogonal, so Put $u_k = v_k / \|v_k\|$, then $\{u_1, \cdots, u_n\}$ is an orthonormal. Clearly, $Q \overset{def}{=} (u_1 \ \cdots \ u_n)$ is unitary.

Moreover, for each $1 \le k \le n$,

$$w_k = \|v_k\| \cdot u_k + \sum_{i=1}^{k-1} \langle w_k, u_i \rangle \cdot u_i \quad \left(= v_k + \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} \cdot v_i \right)$$

Define

$$R_{jk} = \begin{cases} \|v_k\| & j = k \\ \langle w_k, u_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then, R is upper-triangular, and satisfying

$$Q^{-1}A = Q^* A = \begin{pmatrix} \overline{u_1} \\ \vdots \\ \overline{u_n} \end{pmatrix} \cdot (w_1 \ \cdots \ w_n) = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_2, u_1 \rangle & \cdots & \langle w_n, u_1 \rangle \\ \langle w_1, u_2 \rangle & & \ddots & \vdots \\ \vdots & & & \\ \langle w_1, u_n \rangle & \cdots & & \langle w_n, u_n \rangle \end{pmatrix}$$

$$= R.$$

So, $A = QR = QDR'$

where $D = \begin{pmatrix} \|v_1\| & & 0 \\ & \|v_2\| & \\ 0 & & \ddots \\ & & & \|v_n\| \end{pmatrix}$ and $R' = \begin{pmatrix} 1 & & & \\ & 1 & \frac{\langle w_k, u_j \rangle}{\|v_j\|}_{jk} & \\ & & 1 & \\ 0 & & & \ddots \\ & & & & 1 \end{pmatrix}$